





Computer Aided Manufacturing

Laurea Magistrale (Master of Science) in
Automation and Control Engineering

1 year, 1 semester



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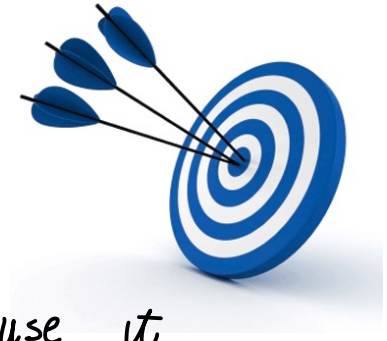
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LEARNING TARGETS

^{Basic class}
This course introduces the ^{manufacturing technologies} student to the fundamentals of manufacturing processes used in discrete-part fabrication along with their needs in terms of automation and computer-based intelligence, according to the smart manufacturing paradigm.



Basic Manufacturing → Needed in the course because it is one of the sector of AUTOMATION

Main Topics:

1. Introduction to Manufacturing
2. Industrial Geometrical Metrology
3. Manufacturing Processes → *Issues related to Processes (Knowledge of requirements Needed)*
4. CNC Machine Tools and Computer-Aided Manufacturing

"Made In Italy"



4 Main sectors of production

in Italian

"LE 4 A"

"4 A": {
AUTOMATION ^{biggest sector in manufacturing}
ALIMENTARE (Food and Beverage)
ABBIGLIAMENTO (Fashion)
ARREDAMENTO (Furniture)

} these sectors from the economical point of view are relevant in ITALY with big gain

MANUFACTURING

→ We need to know

PROBLEMS ?

IT, Automation solve Manufacturing Problems



Lectures and other didactic activities will allow students to:

- know the **structure and behaviour of engineering materials**; *from materials I can know Process*
- understand the **basic design requirements** of a product, and in particular **geometrical product specification and verification**; *"QUALITY" inside requirement (SUSTAINABILITY!)*
- know the **basics of manufacturing and transformation theory**;
- understand the **classification of manufacturing processes** and systems; *METAL & POLYMERS*
- know the **basic principles, tools, and machines** of the considered processes;
- **design feasible manufacturing and inspection processes** for a product;
- know the **computer numerical control (CNC)** and the **CNC machines**; *Automation Applied by machine tools*
- understand the needs and the benefits of the adoption of **CAD/CAM/CAPP techniques**; *STEPS of the process, flow of materials in the system: the sequence of operation must be defined to produce the product*
- understand the **need for automation and computer-based intelligence** in manufacturing.

different kind of resources (device of manufacturing process)

Application of Intelligence on the PROCESS

design of feasible PROCESSES

(PROCESS PLANNING)

transform Materials or Information



LEARNING TARGETS

With respect to the study programme, the course will allow to reach the following generic learning outcomes

(according to the Dublin Descriptors): PERFORMANCE INDICATOR ↗ In the different topics

- IMPORTANT STEPS of Learning {
- capability of knowledge and understanding,
 - capability of applying knowledge and understanding,
 - capability of making judgements, (to choose a seq. of steps, able to define the PROCESS PLAN) → reach a solution
 - capability of communication.

The students should have knowledge related to:

- calculus and geometry; PRE-REQUIREMENTS!
- the basics of physics and mechanics;
- the basics of measurements;
- the basics of automation and control engineering.

We consider a Real Industrial problem → Modelling with abstraction
using a list of hypothesis → if TRUE, I can use the model! { how it is linked to Reality?
(limitation of the model?)



The final grading will be based on **a written test and an oral examination.**



The **written test** will consist of: ↘

"open Book" → List of formula given

- Four numerical problems to be solved in 2 hours aimed at highlighting the student's ability to reason and apply knowledge on simple industrial examples (capability of applying knowledge and understanding). The MS Forms will be used to report numerical results and the grade will be based on these numerical results.

look to
the
numbers →
only :-

↓ passing the WRITTEN TEST

The **oral examination** will consist of: →

able to link the Model to
Reality and describing the formula!

- open questions on the basic topics of the course (capability of knowledge and understanding, capability of communication);
- open questions aimed at highlighting the student's ability to reason and develop links among the different topics of the course (capability of applying knowledge and understanding, of making judgements, and of communication).

MIDDLE TEST in November!



Only for the **first exam** of the winter session of the A.Y., the written test will be divided into an intermediate written test (midterm, November) and a final written test (first exam date, January).

Both **intermediate and final written test** will consist of *three numerical problems to be solved in 1,5 hours* aimed at highlighting the student's ability to reason and apply knowledge on simple industrial examples (capability of applying knowledge and understanding). The MS Forms will be used to report numerical results and the **grade** will be **based on these numerical results**.

The final written test can only be taken if the midterm grade is sufficient. If the student fails the midterm, he/she can take the full exam from the earliest available date in January.

If the student **passes both the intermediate and final written test**, he/she **must take the final oral exam during the first exam session** (that of the final written test).



LEARNING MATERIAL

Mikell P. Groover

Principles of Modern Manufacturing: Materials, Processes, and Systems

John Wiley and Sons, 2016

Geometrical Metrology, CNC based

Robert J. Hocken & Paulo H. Pereira

Coordinate Measuring Machines and Systems

CRC Press, 2012

Mikell P. Groover

Automation, Production Systems, and Computer-Integrated Manufacturing

Pearson, 2015, Chapter 6 and Chapter 7

David Dornfeld and Dae-Eun Lee

Precision Manufacturing

Springer, 2008, eBook, Chapter IX

WeBeep Website: You are responsible for checking the class website at least on a weekly basis.



CLASS ORGANIZATION

Instructional Mix

- Lectures (~ 70 h)
- Class work (~ 30 h)
- Tutoring (20 h) → $\left. \begin{array}{l} \text{In the second part of semester} \\ \text{(IN the Machining PARTS)} \rightarrow \text{solve Numerical example} \\ \text{to Help for Issues} \end{array} \right\} \approx 100h \text{ of Lessons}$

Class Organization:

- in presence (face mask recommended)
- no streaming (streaming will be available on September 23 and 26)
- recording will always be available

Class Schedule

		BOVISA only
Monday	15:15 – 17:15	B8.0.2
Tuesday	13:15 – 15:15	B8.0.2
Thursday	11:15 – 13:15	L.03
Friday	13:15 – 15:15	L.15

